

Continuity of Constraint, Teleportation, and Identity: Limits of Translocation under Bidirectional Constraint Closure

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Abstract

Teleportation, mind-uploading, and nonlocal translocation are recurrent themes in both speculative physics and public imagination. While often dismissed as impossible or accepted uncritically, these concepts demand careful ontological treatment. Building on Bidirectional Constraint Closure (BCC) and the reformulation of Dimension-W as a

Reflective Interface, this paper formalizes the **Continuity of Constraint Theorem (CCT)**, which establishes necessary conditions for identity-preserving coherence. We compare BCC-based limits on teleportation with constraints arising from special relativity and the quantum no-cloning theorem, demonstrating deep structural alignment. We then analyze mind-uploading under the same framework, showing why pattern copying does not imply identity continuity. Finally, a public-facing synthesis explains why most naïve teleportation concepts would result in functional death rather than travel. The analysis preserves empirical rigor while clarifying philosophical and ethical implications.

1. Introduction

Advances in physics, neuroscience, and

computation have revived interest in teleportation and mind-uploading as plausible future technologies. However, discussions often conflate information transfer with identity preservation, and spatial relocation with coherent continuity. This confusion mirrors earlier debates surrounding consciousness, emergence, and personal identity.

The Bidirectional Constraint Closure (BCC) framework provides a principled way to address these issues. By treating reality as stabilized by mirrored constraint regimes whose balance forms a Reflective Interface, BCC reframes motion, coherence, and persistence relationally rather than spatially. This paper extends BCC to formalize constraints on teleportation and identity.

2. The Continuity of Constraint Theorem

(CCT)

2.1 Statement of the Theorem

Continuity of Constraint Theorem (CCT):

A coherent system that stabilizes at the Reflective Interface cannot preserve identity unless its constraint anchoring remains continuous across all transitions. Discontinuous constraint anchoring results in decoherence, collapse of recursive structure, and loss of identity, even if informational configuration is later reconstructed.

2.2 Formal Expression

Let $S(t)$ represent a coherent system at time t , and $C(t)$ its active constraint configuration at the Reflective Interface.

Identity preservation requires:

For all $t_1 < t_2$:

$C(t_1) \rightarrow C(t_2)$ is continuous

If there exists Δt such that $C(t)$ is undefined or discontinuous over Δt , then:

Identity(S) $\neq$ preserved

This condition applies regardless of spatial displacement or informational redundancy.

3. Teleportation under BCC

3.1 Classical Teleportation Assumptions

Popular teleportation models assume:

- Instantaneous spatial relocation
- No interaction with intermediate regions

- Perfect reconstruction from information

Under BCC, these assumptions violate CCT by introducing discontinuous constraint anchoring.

3.2 Comparison with Relativity

Special relativity forbids superluminal signaling to preserve causal order (Einstein, 1905). BCC complements this by forbidding *super-constraint discontinuity*.

Where relativity constrains velocity ($v \leq c$), BCC constrains coherence:

Constraint continuity $\leq$ maximum renegotiation rate

Both frameworks enforce locality, but at different ontological layers.

3.3 Comparison with the Quantum No-Cloning Theorem

The no-cloning theorem states that an arbitrary quantum state $|\psi\rangle$ cannot be copied perfectly (Wootters & Zurek, 1982).

BCC generalizes this principle:

- Quantum no-cloning forbids state duplication
- CCT forbids *identity duplication*

Even perfect classical copying of brain states does not preserve continuity of the Reflective Interface.

4. Why Quantum Teleportation Is Not a Counterexample

Quantum teleportation transfers a quantum state using entanglement and classical communication, but:

- The original state is destroyed
- No conscious identity is involved

Thus, quantum teleportation preserves configuration but not continuity, fully consistent with CCT.

5. Mind-Uploading under BCC

5.1 Pattern Copying vs. Identity Preservation

Mind-uploading proposals assume that replicating neural structure reproduces the self. Under BCC, this confuses pattern similarity with constraint continuity.

Let B represent a biological brain and U an uploaded replica.

Even if:

$\text{Structure}(B) = \text{Structure}(U)$

If:

$C_B(t) \neq$ continuously mapped to $C_U(t)$

Then:

$\text{Identity}(B) \neq \text{Identity}(U)$

The upload is a new coherent system, not the original person.

5.2 Gradual Replacement Scenarios

Even gradual neuron-by-neuron replacement fails unless constraint anchoring remains uninterrupted. The decisive factor is not speed, but continuity of recursive stabilization.

6. Ethical and Practical Implications

- Teleportation that destroys and reconstructs is lethal to the original
- Mind-uploading creates descendants, not survival
- Conscious systems demand continuity-based ethics

These implications arise structurally, not metaphysically.

7. Public Synthesis: Why Teleportation Would Probably Kill You

Teleportation usually imagines copying you somewhere else. But what makes *you* you is not your information alone — it is the uninterrupted balance that holds your experience together.

Break that balance, even for an instant,

and the experience ends.

A perfect copy might wake up elsewhere, believing it is you. But from your point of view, there would be nothing.

Teleportation wouldn't move you. It would replace you.

8. Falsifiability

The framework is falsified if:

- Conscious identity is shown to survive discontinuous constraint collapse
- Identity persists through total annihilation and reconstruction
- Continuity proves irrelevant to experience

No empirical evidence currently supports these conditions.

9. Conclusion

Bidirectional Constraint Closure and the Continuity of Constraint Theorem jointly demonstrate that teleportation and mind-uploading are constrained not by technological limits alone, but by the structure of coherence itself. These constraints align with relativity and quantum mechanics while extending them into the domain of identity and consciousness. Reality permits remarkable flexibility, but it does not permit skipping the conditions that allow experience to exist.

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